

Sihwa Lee

Data Scientist | AI Research | Statistics | Data Engineering

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EDUCATION

University of Seoul

M.S. in Statistics and Data Science

Sep. 2023 – Aug. 2025

Seoul, South Korea

- GPA: 4.06 / 4.50.
- Thesis: *MISA: Mutual Information-driven Separator with Spectral Alignment*.
- Research focus: Domain Generalization, Large Language Models, and Machine Learning.

Hankuk University of Foreign Studies, Global Campus

B.A. in Hungarian Studies; B.A. in Computer and Electronic Systems Engineering

Mar. 2018 – Aug. 2023

Yongin, South Korea

- GPA: 3.83 / 4.50.
- Undergraduate thesis: *Exploring Key Variables in Customer Segmentation through Analysis*.

PUBLICATIONS & PRESENTATIONS

MISA: Mutual Information-driven Separator with Spectral Alignment | 1st Author, Thesis / Preprint 2026

- Proposed a feature-disentanglement framework combining mutual-information manipulation with Gabor- and Sobel-based spectral alignment for domain generalization.
- Evaluated on PACS, VLCS, Office-Home, and DomainNet under the DomainBed protocol and reported competitive or superior results against representative DG baselines.

Multiclass Classification of Online Comments Based on Toxicity Scale | Co-author 2022

- Designed a six-level toxicity scale, built a dataset of **67,434** Korean news comments, and trained a KcBERT-based six-class classifier with **0.89 F1**.
- Published through the Data Youth Campus project in *Advances in Computer Science and Ubiquitous Computing*.

Exploring Key Variables in Customer Segmentation through Analysis | 1st Author, B.S. Thesis 2022

- Combined clustering, classification, PCA, and factor analysis on customer purchase data to identify income, purchase amount, purchase frequency, and number of children as key segmentation variables.
- Compared K-means, Gaussian mixture models, and mean-shift clustering; selected mean-shift with a silhouette coefficient of **0.6764**.

EXPERIENCE

NC AI

AI Researcher, Data Engineering Team

Sep. 2025 – Mar. 2026

South Korea

- Contributed to building a multilingual pretraining text dataset (**150M sentences**) for an independent from-scratch foundation-model project, including large-scale cleaning of Chinese/Japanese Wikipedia data.
- Analyzed externally sourced post-training data for duplication rate, domain/language diversity, and key statistics; established validation criteria and drove quality remediation with outsourcing partners, supporting **NIA eligibility approval** and AI Hub publication.
- Built an image-and-JSON fashion dataset (**10,000 images**) from web-crawling outputs, designing the schema/mapping pipeline and implementing automated error-detection code to reduce manual review burden.
- Generated and validated e-commerce detail-page contents against business requirements, designing test cases and tuning prompts/generation conditions to improve accuracy and consistency for **VAETKI Commerce**.
- Developed a story-driven game prototype **end-to-end** using Streamlit and the OpenAI API, including generative dialogue-automation logic and UX design.

PROJECTS

Foundation Model Pretraining Data Pipeline | *World Best LLM, NC AI* Sep. 2025 – Jan. 2026

- Implemented document-cleaning pipelines for URL, comment, table/formula, and markup-like artifacts; aligned interfaces with upstream and downstream preprocessing steps to reduce pipeline conflicts.
- Summarized quality metrics and failure cases into reports that could be used by non-research stakeholders for review, improvement requests, and final validation.

Fin:AI Challenge | *JB Financial Group Hackathon* Jun. 2026

- Prepared an MVP proposal and functional specification for an AI-enabled financial service, emphasizing evidence presentation, approval flow, regulatory traceability, and post-hoc auditability.

Market Sales Analytics Dashboard | *University of Seoul* Oct. 2023 – Dec. 2023

- Independently built an interactive R Shiny dashboard analyzing sales, customer, and product/delivery data.
- Performed RFM-based K-means clustering for customer segmentation and Kruskal-Wallis testing for category-level sales comparison.

Data Youth Campus Natural Language Processing Project | *HUFS* Jun. 2022 – Aug. 2022

- Crawled and labeled Naver News comments across social, economic, and cultural domains; defined Positive, Neutral, Negative, Slur, Violent, and Hate classes.
- Built an end-to-end NLP pipeline covering data collection, labeling criteria, augmentation, preprocessing, KcBERT fine-tuning, evaluation, and paper writing.

TEACHING

Teaching Assistant, Department of Statistics and Data Science, University of Seoul Mar. 2024 – Jun. 2025

- **Deep Learning** (Mar. 2024 – Jun. 2024)
- **Advanced Statistical Programming** (Sep. 2024 – Dec. 2024)
- **Regression Analysis I** (Mar. 2025 – Jun. 2025)
- **Introduction to Big Data II** (Mar. 2025 – Jun. 2025)

Coordinator, Department of Statistics and Data Science, University of Seoul Sep. 2024 – Oct. 2024

- **Data Analysis Competition**

AWARDS

Creativity Award, 2024 K-Data Science Hackathon Nov. 2024

- Awarded by the Ministry of Science and ICT, National Research Foundation of Korea, and University of Seoul ALSH for an active-learning framework validating synthetic medical data utility.

CERTIFICATIONS

SQL Developer (SQLD), Korea Data Agency Jun. 2026

Google Analytics Certification, Google May. 2026

TOEIC Speaking IH, ETS Mar. 2025

Advanced Data Analytics Semi-Professional (ADsP), Korea Data Agency Nov. 2022

SKILLS

Programming : Python, MySQL, PostgreSQL, R, JavaScript

Machine Learning : PyTorch, Scikit-learn, Pandas, NumPy, BigQuery

Tools : Git, LaTeX, Overleaf, Jupyter, Google Colab